

Autofs

AutoFS (Automatic File System) is a Linux service that **automatically mounts** a remote filesystem (such as an NFS share) **only when you access it**. When you stop using it for some time, AutoFS **automatically unmounts** it.

NFS Mount

Mounts immediately after boot

Always mounted

Uses system resources continuously

Manual unmount

AutoFS

Mounts only when accessed

Mounted only when needed

Saves resources

Automatic unmount

Server side :

Step1

```
[root@server ~]# cd /
[root@server ~]# mkdir /nfs_share
[root@server ~]# cd /
[root@server /]# ls
afs  bin  boot  dev  etc  home  lib  lib64  media  mnt  nfs_share
```

```
[root@server ~]# touch /nfs_share/file{1..5}
```

Step2

vi = Open file editor

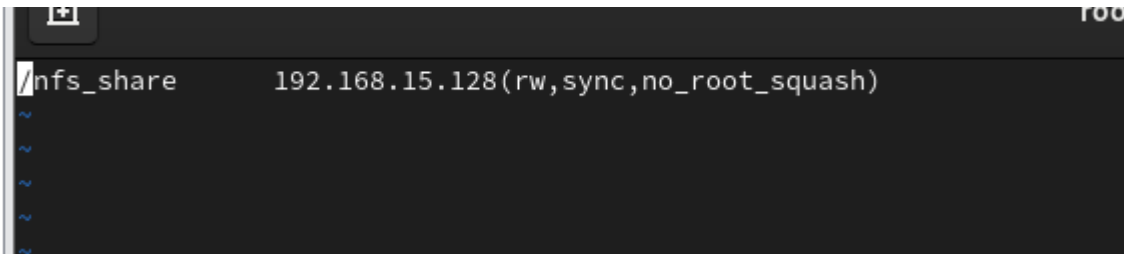
/etc/exports = NFS configuration file

Rw -Read and Write permission.

Write data immediately.

Client root user remains root.

```
[root@server ~]# vi /etc/exports
```



Step3

Restart the NFS service.

Start NFS automatically after every reboot.

```
[root@server ~]# systemctl restart nfs-server.service
[root@server ~]# systemctl status nfs-server.service
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: disabled)
   Drop-In: /run/systemd/generator/nfs-server.service.d
            └─order-with-mounts.conf
   Active: active (exited) since Sat 2026-06-27 10:56:25 IST; 11s ago
     Docs: man:rpc.nfsd(8)
           man:exportfs(8)
  Process: 4261 ExecStartPre=/usr/sbin/exportfs -r (code=exited, status=0/SUCCESS)
  Process: 4262 ExecStart=/usr/sbin/rpc.nfsd (code=exited, status=0/SUCCESS)
  Process: 4272 ExecStart=/bin/sh -c if systemctl -q is-active gssproxy; then systemctl reload gssproxy
 Main PID: 4272 (code=exited, status=0/SUCCESS)
    CPU: 114ms
```

Step4

```
Jun 27 10:56:25 server systemd[1]: Finished NFS ser  
[root@server ~]# systemctl disable firewalld
```

Step5

```
[root@server ~]# getenforce  
Enforcing
```

```
[root@server ~]# setenforce 0  
[root@server ~]# getenforce  
Permissive
```

```
[root@client ~]# yum install autofs  
Last metadata expiration check: 0:11:49 ago on Sat 27 Jun 2026 11:06:56 AM IST.  
Package autofs-1:5.1.7-68.el9.x86_64 is already installed.  
Dependencies resolved.  
Nothing to do.  
Complete!
```

Then I go client

Step1

Step2

Open AutoFS master configuration file.

```
[root@client ~]# vi /etc/auto.master
```

server

```
root@client:~# vi /etc/auto.master
#
# Sample auto.master file
# This is a 'master' automounter map and it has the following format:
# mount-point [map-type[,format]:]map [options]
# For details of the format look at auto.master(5).
#
/misc      /etc/auto.misc
/car       /etc/auto.misc
#
# NOTE: mounts done from a hosts map will be mounted with the
#       "nosuid" and "nodev" options unless the "suid" and "dev"
#       options are explicitly given.
#
/net       -hosts
#
# Include /etc/auto.master.d/*.autoofs
# To add an extra map using this mechanism you will need to add
# two configuration items - one /etc/auto.master.d/extra-autoofs file
```

Step3

Create a configuration file that contains remote NFS information.

```
[root@client ~]# vi /etc/auto.master
[root@client ~]# vi /etc/auto.misc
[root@client ~]# systemctl restart autofs.service
```

client

```
# This is an automounter map and it has the following format
# key [ -mount-options-separated-by-comma ] location
# Details may be found in the autoofs(5) manpage

cd          -fstype=iso9660,ro,nosuid,nodev :/dev/cdrom

# the following entries are samples to pique your imagination
#linux      -ro,soft          ftp.example.org:/pub/linux
#boot       -fstype=ext2      :/dev/hda1
#floppy      -fstype=auto      :/dev/fd0
#floppy      -fstype=ext2      :/dev/fd0
#e2floppy    -fstype=ext2      :/dev/fd0
#jaz         -fstype=ext2      :/dev/sdc1
#removable   -fstype=ext2      :/dev/hdd
small       -fstype=nfs        192.168.15.130:/nfs_share
~
~
```

Step4

Restart AutoFS so it reads the new configuration.

```
[root@client ~]# vi /etc/auto.misc
[root@client ~]# systemctl restart autofs.service
```

Step5

```
[root@client ~]# systemctl restart autofs.service  
[root@client ~]# systemctl enable --now autofs.service
```

Step6

```
[root@client ~]# cd /car  
[root@client car]# cd small  
[root@client small]# ls  
file1 file2 file3 file4 file5
```

Step7

```
[root@client small]# pwd  
/car/small  
[root@client small]# ll  
total 0  
-rw-r--r--. 1 root root 0 Jun 27 11:03 file1  
-rw-r--r--. 1 root root 0 Jun 27 11:03 file2  
-rw-r--r--. 1 root root 0 Jun 27 11:03 file3  
-rw-r--r--. 1 root root 0 Jun 27 11:03 file4  
-rw-r--r--. 1 root root 0 Jun 27 11:03 file5
```

Step8

Human readable.

```
[root@client small]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	1.3G	0	1.3G	0%	/dev
tmpfs	1.3G	0	1.3G	0%	/dev/shm
tmpfs	522M	8.0M	514M	2%	/run
/dev/mapper/cs-root	27G	5.4G	21G	21%	/
/dev/sda1	960M	527M	434M	55%	/boot
tmpfs	261M	96K	261M	1%	/run/user/1000
/dev/sr1	15G	15G	0	100%	/run/media/raj/CentOS-Stream-9-BaseOS-x86_64
192.168.15.130:/nfs_share	27G	6.2G	20G	24%	/car/small

Then

Two type to change the timeout (default timeout =300 sec or 5min)

1.global

2.specific

1.

```
[root@client ~]# vi /etc/autofs.conf
```

server

```
root@client
#
# Global configuration options.
#
# open_file_limit - set the maximum number of open files. Note there
#                   may be other limits within the system that prevent
#                   this from being set, systemd for example may need
#                   a setting in the unit file to increase its default.
#                   The autofs default is 20480.
#
#open_file_limit = 20480
#
# Define default options for autofs.
#
[ autofs ]
#
# master_map_name - default map name for the master map.
#
#master_map_name = auto.master
#
# timeout - set the default mount timeout in seconds. The internal
#            program default is 10 minutes, but the default installed
#            configuration overrides this and sets the timeout to 5
#            minutes to be consistent with earlier autofs releases.
#
timeout = 300
#
```

2. specific

```
[root@client ~]# vi /etc/auto.master
```

SEI VCI

```
#
# Sample auto.master file
# This is a 'master' automounter map and it has the following format:
# mount-point [map-type[,format]:]map [options]
# For details of the format look at auto.master(5).
#
/misc    /etc/auto.misc
/car     /etc/auto.misc --timeout=60
#
# NOTE: mounts done from a hosts map will be mounted with the
#       "nosuid" and "nodev" options unless the "suid" and "dev"
#       options are explicitly given.
#
/net     -hosts
#
```